BLOCKCHAIN TECHNOLOGY

INTRODUCTION

UNIT - I

Introduction to Blockchain Technology:

What is Blockchain?

A blockchain is a constantly growing ledger which keeps a permanent record of all the transactions that have taken place in a secure, chronological, and immutable way. Let's breakdown the definition,

- **Ledger:** It is a file that is constantly growing.
- **Permanent:** It means once the transaction goes inside a blockchain, you can put up it permanently in the ledger.
- **Secure:** Blockchain placed information in a secure way. It uses very advanced cryptography to make sure that the information is locked inside the blockchain.
- **Chronological:** Chronological means every transaction happens after the previous one.
- **Immutable:** It means as you build all the transaction onto the blockchain, this ledger can never be changed.

A blockchain is a chain of blocks which contain information. Each block records all of the recent transactions, and once completed goes into the blockchain as a permanent database. Each time a block gets completed, a new block is generated.

Blockchain is a buzzword in today's technology and this technology is described as the most disruptive technology of the decade. Thus, Blockchain is used for the secure transference of items like money, contracts, property rights, stocks, and even networks without any requirement of Third Party Intermediaries like Governments, banks, etc. Once the data is stored in the Blockchain it becomes very difficult to manipulate the stored data. A Blockchain is a Network Protocol like SMTP. However, Blockchain cannot be run without the Internet. BlockChain is useful in many areas like Banking, Finance, Healthcare, Insurance, etc.

A blockchain is an open, distributed ledger that can record transactions between two parties efficiently and in averifiable and permanent way without the need for a central authority.

Key Characteristics:

- **Open**: Anyone can access blockchain.

- **Distributed or Decentralized:** Not under the control of any single authority.
- **Efficient:** Fast and Scalable.
- **Verifiable:** Everyone can check the validity of information because each node maintains a copyof the transactions.
- **Permanent:** Once a transaction is done, it is persistent and can't be altered.

Blockchain can be defined as the Chain of Blocks that contain some specific Information. Thus, a Blockchain is a ledger i.e file that constantly grows and keeps the record of all transactions permanently. This process takes place in a secure, chronological (Chronological means every transaction happens after the previous one) and immutable way. Each time when a block is completed in storing information, a new block is generated.

Who uses the blockchain?

Blockchain technology can be integrated into multiple areas. The primary use of blockchains is as a distributed ledger for cryptocurrencies. It shows great promise across a wide range of business applications like Banking, Finance, Government, Healthcare, Insurance, Media and Entertainment, Retail,etc.

Need of Blockchain:

Blockchain technology has become popular because of the following.

- **Time reduction:** In the financial industry, blockchain can allow the quicker settlement of trades. It does not take a lengthy process for verification, settlement, and clearance. It is because of a single version of agreed-upon data available between all stakeholders.

- **Unchangeable transactions:** Blockchain register transactions in a chronological order which certifies the unalterability of all operations, means when a new block is added to the chain of ledgers, it cannot be removed or modified.
- **Reliability:** Blockchain certifies and verifies the identities of each interested parties. This removes double records, reducing rates and accelerates transactions.
- **Security:** Blockchain uses very advanced cryptography to make sure that the information is locked inside the blockchain. It uses Distributed Ledger Technology where each party holds a copy of the original chain, so the system remains operative, even the large number of other nodes fall.
- **Collaboration:** It allows each party to transact directly with each other without requiring a third-party intermediary.
- **Decentralized:** It is decentralized because there is no central authority supervising anything. There are standards rules on how every node exchanges the blockchain information. This method ensures that all transactions are validated, and all valid transactions are added one by one.

Distributed Systems:

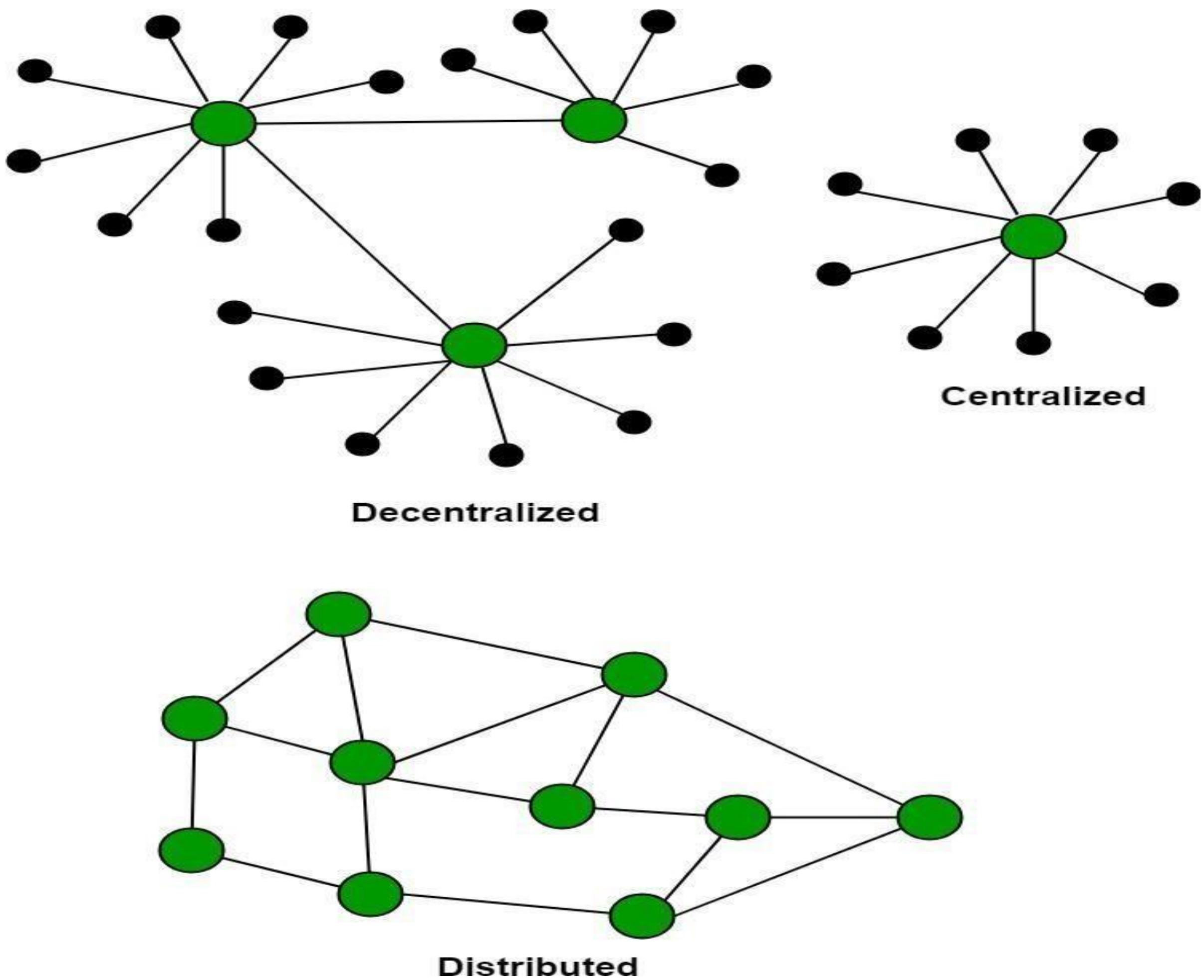
Understanding distributed systems is essential to our understanding blockchain, as blockchain was a distributed system at its core. It is a distributed ledger that can be centralized or decentralized. A blockchain is originally intended to be and is usually used as a decentralized platform. It can be thought ofas a system that has propertiesof the both decentralized and distributed paradigms. It is a decentralized- distributed system.

Distributed systems are a computing paradigm whereby two or more nodes work with each other in a coordinated fashion to achieve a common outcome. It is modeled in such a way that end users see it as a single logical platform. For example, Google's search engine is based on a large distributed system; however, to a user, it looks like a single, coherent platform.

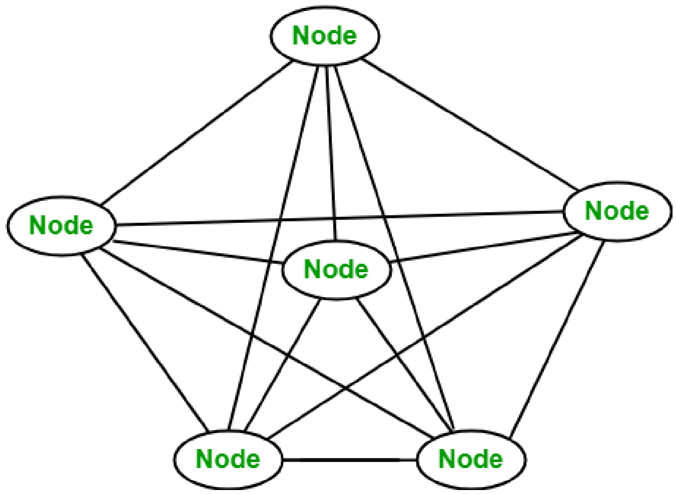
A **node** can be defined as an individual player in a distributed system. All nodes are capable of sending and receiving messages to and from each other. There is no Central Server or System which keeps the dataof Blockchain. The data is distributed over Millions of Computers around the world which are connected with the Blockchain. This system allows Notarization of Data as it is present on every Node and is publicly verifiable.A node can be defined as an individual player in a distributed system. All nodes are capable of sending and receiving messages to and from each other.

Nodes can be honest, faulty, or malicious and have their own memory and processor. A node that can exhibit arbitrary behavior is also known as a Byzantine node. This arbitrary behavior can be intentionally malicious, which is detrimental to the operation of the network. Generally, any unexpected behavior of a node on the network can be categorized as Byzantine. This term arbitrarily encompasses any behavior that is unexpected or malicious.

The main challenge in distributed system design is coordination between nodes and fault tolerance. Even if someof the nodes become faulty or network links break, the distributed system should tolerate this and should continue to work flawlessly in order to achieve the desired result. This has been an area of active research for many years and several algorithms and mechanisms have been proposed to overcome these issues.



A network of nodes: A node is a computer connected to the Blockchain Network. Node gets connected with Blockchain using the client. Client helps in validating and propagates transaction on to the Blockchain. When a computer connects to the Blockchain, a copy of the Blockchain data gets downloaded into the system and the node comes in sync with the latest block of data on Blockchain. The Node connected to the Blockchain which helps in the execution of a Transaction in return for an incentive is called Miners.



Disadvantages of current transaction system:

 Cash can only be used in low amount transaction locally.

 Huge waiting time in the processing of transactions.

 Need to third party for verification and execution of Transaction make the process complex.

 If the Central Server like Banks is compromised, whole System is affected including the participants.

 Organization doing validation charge high process thus making the process expensive.

Building trust with Blockchain:

Blockchain enhances trust across a business network. It's not that you can't trust those who you conduct business with its that you don't need to when operating on a Blockchain network.

Blockchain builts trust through the following five attributes:

 Distributed: The distributed ledger is shared and updated with every incoming transaction among the nodes connected to the Blockchain. All this is done in real-time as there is no central server controlling thedata.

 Secure: There is no unauthorized access to Blockchain made possible through Permissions and Cryptography.

 Transparent: Because every node or participant in Blockchain has a copy of the Blockchain data, theyhave access to all transaction data. They themselves can verify the identities without the need for mediators.

 Consensus-based: All relevant network participants must agree that a transaction is valid. This is achieved through the use of consensus algorithms.

 Flexible: Smart Contracts which are executed based on certain conditions can be written into the platform. Blockchain Network can evolve in pace with business processes.

History of Blockchain:

- In 1991, researcher scientists named Stuart Haber and W. Scott Stornetta introduce Blockchain Technology. These scientists wanted some Computational practical Solution for time-stamping the digital documents so that they couldn't be tempered or misdated. So both scientists together developed a system with the help of Cryptography. In this System, the time-stamped documents are stored in a Chain of Blocks.
- After that in 1992, Merkle Trees formed a legal corporation by using a system developed by Stuart Haber and W. Scott Stornetta with some more features. Hence, Blockchain Technology became efficient to store several documents to be collected into one block. Merkle used a Secured Chain of Block which storesmultiple data records in a sequence. However, this Technology became unused when Patent came into existence in 2004.
- However, in the same year 2004, Cryptographic activist Hal Finney introduced a system for digital cash known as "Reusable Proof of Work". This step was the game-changer in the history of Blockchainand Cryptography. This System helps others to solve the Double Spending Problem by keeping the ownership of tokens registered on a trusted server.
- After that in 2008, Satoshi Nakamoto conceptualized the concept of "Distributed Blockchain" under his white paper: "A Peer to Peer Electronic Cash System". He modified the model of Merkle Tree and created a system that is more secure and contains the secure history of data exchange. His System follows a peer-to-peer network of time stamping. His system became so useful that Blockchain become the backbone of Cryptography.
- After that, the evolution of Blockchain is steady and promising and became a need in various fields. Blockchain technology is so secure that the following surprising news will give proof about that. A person named, James Howells was an IT worker in the United Kingdom, he starts mining bitcoins which are partof Blockchain in 2009 and stopped this in 2013. He spends $17,000 on it and after he stopped he sells the parts of his laptop on eBay and keep the drive with him so that when he needs to work again on bitcoinhe will utilize it but while cleaning his house in 2013, he thrashed his drive with garbage and now his bitcoins cost nearly $127 million. This money now remains unclaimed in the Bitcoin system.

The blockchain is the public ledger of all Bitcoin transactions that have ever been exe- cuted. It is constantly growing as miners add new blocks to it (every 10 minutes) to record the most recent transactions. The blocks are added to the blockchain in a lin- ear, chronological order. Each full node (i.e., every computer connected to the Bitcoin network using a client that performs the task of validating and relaying transactions) has a copy of the blockchain, which is downloaded automatically when the miner joins the Bitcoin network. The blockchain has complete information about addresses and balances from the genesis block (the very first transactions ever executed) to the most recently completed block.

Blockchain is the backbone Technology of Digital CryptoCurrency BitCoin. The blockchain is a distributed database of records of all transactions or digital event that have been executed and shared among participating parties. Each transaction verified by the majority of participants of the system. It contains every single record of each transaction. BitCoin is the most popular cryptocurrency an example of the blockchain. Blockchain Technology Records Transaction in Digital Ledger which is distributedover the Network thus making it incorruptible. Anything of value like Land Assets, Cars, etc. can be recorded on Blockchain as a Transaction.

One of the famous use of Blockchain is Bitcoin. The bitcoin is a cryptocurrency and is used to exchange digital assets online. Bitcoin uses cryptographic proof instead of third-party trust for two parties to execute transactions over the internet. Each transaction protects through digital signature.

CAP theorem:

The **CAP theorem**, also known as Brewer's theorem, was introduced by Eric Brewer in 1998 as a conjecture. In 2002, it was proven as a theorem by Seth Gilbert and Nancy Lynch. The theorem states that any distributed system cannot have consistency, availability, and partition tolerance simultaneously:

- **Consistency** is a property that ensures that all nodes in a distributed system have a single, current, and identical copy of the data. Consistency is achieved using consensus algorithms in order to ensure that all nodes have the same copy of the data. This is also called **state machine replication**. The blockchain is a means for achieving state machine replication.
- **Availability** means that the nodes in the system are up, accessible for use, and are accepting incoming requests and responding with data without any failures as and when required. In other words, data is available at each node and the nodes are responding...

The CAP theorem states that **a distributed database system has to make a tradeoff between Consistency and Availability when a Partition occurs**. A distributed database system is bound to have partitions in a real-worldsystem due to network failure or some other reason.

The CAP Theorem is comprised of three components (hence its name) as they relate to distributed datastores:

- **Consistency.** All reads receive the most recent write or an error.

- **Availability.** All reads contain data, but it might not be the most recent.
- **Partition tolerance.** The system continues to operate despite network failures (ie; dropped partitions, slow network connections, or unavailable network connections between nodes.)

In normal operations, your data store provides all three functions. But the CAP theorem maintains that when a distributed database experiences a network failure, you can provide either consistency or availability.

It's a tradeoff. All other times, all three can be provided. But, in the event of a network failure, a choice must be made.In the theorem, partition tolerance is a must. The assumption is that the system operates on a distributed data store so the system, by nature, operates with network partitions. Network failures will happen, so to offer any kind of reliable service, partition tolerance is necessary—the P of CAP.

That leaves a decision between the other two, C and A. When a network failure happens, one can choose to guarantee consistency or availability:

- High consistency comes at the cost of lower availability.
- High availability comes at the cost of lower consistency.

Benefits and limitations of blockchain:

Numerous benefits of blockchain technology are being discussed in the industry and proposed by thought leadersaround the world in blockchain space. The top 10 benefits are listed and discussed as follows.

Decentralization:

This is a core concept and benefit of blockchain. There is no need for a trusted third party or intermediary tovalidate transactions; instead a consensus mechanism is used to agree on the validity of transactions.

Transparency and trust:

As blockchains are shared and everyone can see what is on the blockchain, this allows the system to be transparent and as a result trust is established. This is more relevant in scenarios such as the disbursement offunds or benefits where personal discretion should be restricted.

Immutability:

Once the data has been written to the blockchain, it is extremely difficult to change it back. It is not truly immutable but, due to the fact that changing data is extremely difficult and almost impossible, this is seen as a benefit to maintaining an immutable ledger of transactions.

High availability:

As the system is based on thousands of nodes in a peer-to-peer network, and the data is replicated and updated oneach and every node, the system becomes highly available. Even if nodes leave the network or become inaccessible, the network as a whole continues to work, thus making it highly available.

Highly secure:

All transactions on a blockchain are cryptographically secured and provide integrity.

Simplification of current paradigms:

The current model in many industries such as finance or health is rather disorganized, wherein multiple entities maintain their own databases and data sharing can become very difficult due to the disparate nature of the systems. But as a blockchain can serve as a single shared ledger among interested parties, this can result in simplifying this model by reducing the complexity of managing the separate systems maintained by each entity.

Faster dealings:

In the financial industry, especially in post-trade settlement functions, blockchain can play a vital role by allowing the quicker settlement of trades as it does not require a lengthy process of verification, reconciliation, and clearance because a single version of agreed upon data is already available on a shared ledger between financial organizations.

Cost saving:

As no third party or clearing houses are required in the blockchain model, this can massively eliminate overheadcosts in the form of fees that are paid to clearing houses or trusted third parties.

Decentralization:

Decentralization is a core benefit and service provided by blockchain technology. By design, blockchain is a perfect vehicle for providing a platform that does not need any intermediaries and that can function with many different leaders chosen via consensus mechanisms. This model allows anyone to compete to become the decision-making authority. A consensus mechanism governs this competition, and the most famous method is known as **Proof of Work** (**PoW**).

Decentralization is applied in varying degrees from a semi-decentralized model to a fully decentralized one depending on the requirements and circumstances. Decentralization can be viewed from a blockchain perspective as a mechanism that provides a way to remodel existing applications and paradigms, or to build new applications, to give full control to users.

Information and communication technology (**ICT**) has conventionally been based on a centralized paradigm whereby database or application servers are under the control of a central authority, such as a system administrator. With Bitcoin and the advent of blockchain technology, this model has changed, and now the technology exists to allow anyone to start a decentralized system and operate it with no single point of failure or single trusted authority. It can either be run autonomously or by requiring some human intervention, depending on the type and model of governance used in the decentralized application running on the blockchain.

Different types of networks/systems

Centralized systems are conventional (client-server) IT systems in which there is a single authority that controlsthe system, and who is solely in charge of all operations on the system. All users of a centralized system are dependent on a single source of service. The majority of online service providers, including Google, Amazon, eBay, and Apple's App Store, use this conventional model to deliver services.

In a **distributed system**, data and computation are spread across multiple nodes in the network. Sometimes, this term is confused with *parallel computing*. While there is some overlap in the definition, the main difference between these systems is that in a parallel computing system, computation is performed by all nodes simultaneously in order to achieve the result; for example, parallel computing platforms are used in weather research and forecasting, simulation, and financial modeling. On the other hand, in a distributed system, Computation may not happen in parallel and data is replicated across multiple nodes that users view as a single, coherent system. Variations of both of these models are used to achieve fault tolerance and speed. In the parallel system model, there is still a central authority that has control over all nodes and governs processing. This meansthat the system is still centralized in nature.

The critical difference between a decentralized system and distributed system is that in a distributed system, there is still a central authority that governs the entire system, whereas in a decentralized system, no such authority exists.

A **decentralized system** is a type of network where nodes are not dependent on a single master node; instead, control is distributed among many nodes. This is analogous to a model where each department in an organizationis in charge of its own database server, thus taking away the power from the central server and distributing it to the sub-departments, who manage their own databases.

A significant innovation in the decentralized paradigm that has given rise to this new era of decentralization of applications is **decentralized consensus**. This mechanism came into play with Bitcoin,and it enables a user to agree on something via a consensus algorithm without the need for a central, trusted third party, intermediary, or service provider.

We can also now view the different types of networks shown earlier from a different perspective, where we highlight the controlling authority of these networks as a symbolic hand, as shown in the following diagram. This model provides a clearer understanding of the differences between these networks from a decentralization point of view.

Feature	Centralized	Decentralized
Ownership	Service provider	All users
Architecture	Client/server	Distributed, different topologies
Security	Basic	More secure
High availability	No	Yes
Fault tolerance	Basic, single point of failure	Highly tolerant, as service is replicated
Collusion resistance	Basic, because it's under the control of agroup or even single individual	Highly resistant, as consensus algorithmsensure defense against adversaries
Application architecture	Single application	Application replicated across all nodes onthe network
Trust	Consumers have to trust the service provider	No mutual trust required
Cost for consumer	Higher	Lower

Methods of Decentralization:

Two methods can be used to achieve decentralization: disintermediation and competition. These methods will bediscussed in detail in the sections that follow.

The concept of **disintermediation** can be explained with the aid of an example. Imagine that you want to send money to a friend in another country. You go to a bank, which, for a fee, will transfer your money to the bank in that country. In this case, the bank maintains a central database that is updated, confirming that you have sent the money. With blockchain technology, it is possible to send this money directly to your friend without the need for a bank. All you need is the address of your friend on the blockchain. This way, ~~the intermediary (that is, the bank) is no longer required, and decentralization is achieved by~~

disintermediation. It is debatable, however, how practical decentralization through disintermediation is in the financial sector due to the massive regulatory and compliance requirements. Nevertheless, this model can be used not only in finance but in many other industries as well, such as health, law, and the public sector. In the health industry, where patients, instead of relying on a trusted third party (such as the hospital record system) can be in full control of their own identity and their data that they can share directly with only those entities that they trust. As a general solution, blockchain can serve as a decentralized health record management system where health records can be exchanged securely and directly between different entities (hospitals, pharmaceutical companies, patients) globally without any central authority.

Contest-driven decentralization:

In the method involving **competition**, different service providers compete with each other in order to be selected for the provision of services by the system. This paradigm does not achieve complete decentralization. However, to a certain degree, it ensures that an intermediary or service provider is not monopolizing the service. In the context of blockchain technology, a system can be envisioned in which smart contracts can choose an external data provider from a large number of providers based on their reputation, previous score, reviews, and quality of service.

This method will not result in full decentralization, but it allows smart contracts to make a free choice based on the criteria just mentioned. This way, an environment of competition is cultivated among service providers wherethey compete with each other to become the data provider of choice.

In the following diagram, varying levels of decentralization are shown. On the left side, the conventional approach is shown where a central system is in control; on the right side, complete disintermediation is achieved, as intermediaries are entirely removed. Competing intermediaries or service providers are shown in the center. At that level, intermediaries or service providers are selected based on reputation or voting, thus achieving partialdecentralization:

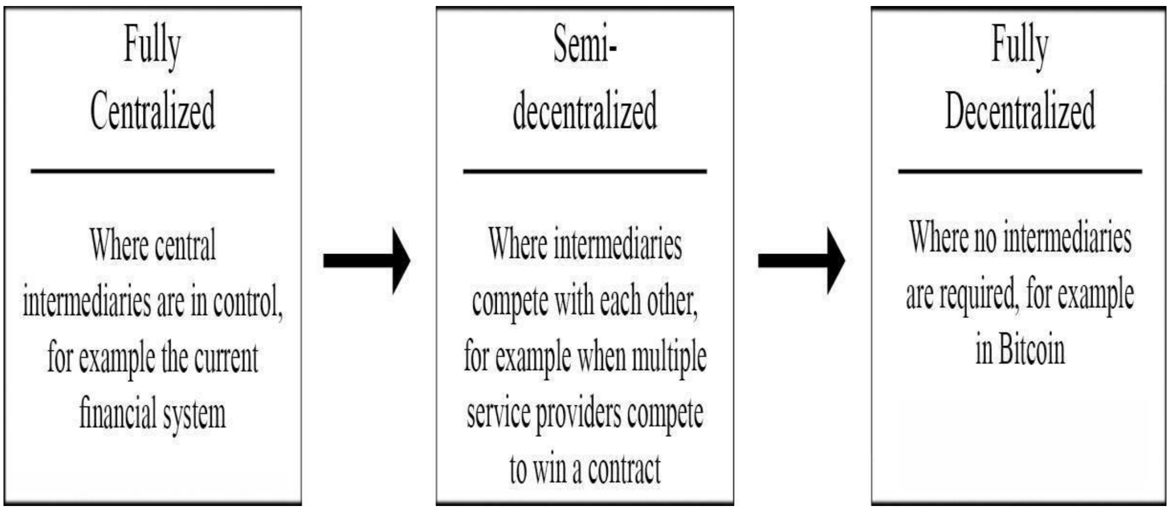


Figure : Scale of decentralization

There are many benefits of decentralization, including transparency, efficiency, cost saving, development of trusted ecosystems, and in some cases privacy and anonymity. Some challenges, such as security requirements, software bugs, and human error, need to be examined thoroughly. This view raises some fundamental questions. Is a blockchain really needed? When is a blockchain required? In what circumstances is blockchain preferable to traditional databases? To answer these questions, go through the simple set of questions presented below:

Question	Yes/No	Recommended solution
Is high data throughput required?	Yes	Use a traditional database.
	No	A central database might still be useful if other requirements are met. For example, if users trust each other, then perhaps there is no need fora blockchain. However, if they don't or trust cannot be established for any reason, blockchain can be helpful.

Are updates centrally controlled?	Yes	Use a traditional database.
	No	You may investigate how a public/private blockchain can help.
Do users trust eachother?	Yes	Use a traditional database.
	No	Use a public blockchain.
Are users anonymous?	Yes	Use a public blockchain.
	No	Use a private blockchain.
Is consensus required tobe maintained within a consortium?	Yes	Use a private blockchain.
	No	Use a public blockchain.
Is strict data immutabilit yrequired?	Yes	Use a blockchain.
	No	Use a central/traditional database.

Answering all of these questions can help you decide whether or not a blockchain is required or suitable for solving the problem. Beyond the questions posed in this model, there are many other issues to consider, such as latency, choice of consensus mechanisms, whether consensus is required or not, and where consensus is going to be achieved. If consensus is maintained internally by a consortium, then a private blockchain should be used; otherwise, if consensus is required publicly among multiple entities, then a public blockchain solution should be considered. Other aspects, such as immutability, should also be considered when deciding whether to use a blockchain or a traditional database. If strict data immutability is required, then a public blockchain should be used; otherwise, a central database may be an option.

As blockchain technology matures, there will be more questions raised regarding this selection model. For now, however, this set of questions is sufficient for deciding whether a blockchain-based solution is suitable or not.

Now we understand different methods of decentralization and have looked at how to decide whether a blockchain is required or not in a particular scenario. Let's now look at the process of decentralization, that is, how we can take an existing system and decentralize it.

Routes to decentralization:

There are systems that pre-date blockchain and Bitcoin, including BitTorrent and the Gnutella file-sharing system, which to a certain degree could be classified as decentralized, but due to a lack of any incentivization mechanism, participation from the community gradually decreased. There wasn't any incentive to keep the users interested in participating in the growth of the network. With the advent of blockchain technology, many initiatives are being taken to leverage this new technology to achieve decentralization. The Bitcoin blockchain is typically the first choice for many, as it has proven to be the most resilient and secure blockchain and has a market cap of nearly $166 billion at the time of writing. Alternatively, other blockchains, such as Ethereum, serve as the tool of choice for many developers for building decentralized applications. Compared to Bitcoin, Ethereum has become a more prominent choice because of the flexibility it allows for programming any business logic into the blockchain by using **smartcontracts**.

How to decentralize

The framework raises four questions whose answers provide a clear understanding of how a system can be decentralized:

1. What is being decentralized?

2. What level of decentralization is required?
3. What blockchain is used?
4. What security mechanism is used?

The first question simply asks you to identify what system is being decentralized. This can be any system, suchas an identity system or a trading system.

The second question asks you to specify the level of decentralization required by examining the scale of decentERalization, as discussed earlier. It can be full disintermediation or partial disintermediation.

The third question asks developers to determine which blockchain is suitable for a particular application. It canbe Bitcoin blockchain, Ethereum blockchain, or any other blockchain that is deemed fit for the specific application.

Finally, a fundamental question that needs to be addressed is how the security of a decentralized system will be guaranteed. For example, the security mechanism can be atomicity-based, where either the transaction executesin full or does not execute at all. This deterministic approach ensures the integrity of the system. Other mechanisms may include one based on reputation, which allows for varying degrees of trust in a system.

In the following section, let's evaluate a money transfer system as an example of an application selected to bedecentralized.

Decentralization framework example:

The four questions discussed previously are used to evaluate the decentralization requirements of this application. The answers to these questions are as follows:

1. Money transfer system
2. Disintermediation
3. Bitcoin
4. Atomicity

The responses indicate that the money transfer system can be decentralized by removing the intermediary, implemented on the Bitcoin blockchain, and that a security guarantee will be provided via atomicity. Atomicity will ensure that transactions execute successfully in full or do not execute at all. We have chosen the Bitcoin blockchain because it is the longest established blockchain and has stood the test of time.

Similarly, this framework can be used for any other system that needs to be evaluated in terms of decentralization. The answers to these four simple questions help clarify what approach to take to decentralizethe system.

To achieve complete decentralization, it is necessary that the environment around the blockchain also be decentralized. We'll look at the full ecosystem of decentralization next.

The blockchain is a distributed ledger that runs on top of conventional systems. These elements include storage,communication, and computation.

Storage

Data can be stored directly in a blockchain, and with this fact it achieves decentralization. However, a significant disadvantage of this approach is that a blockchain is not suitable for storing large amounts of data by design. It can store simple transactions and some arbitrary data, but it is certainly not suitable for storing images or large blobs of data, as is the case with traditional database systems.

A better alternative for storing data is to use **distributed hash tables** (**DHTs**). DHTs were used initially in peer- to-peer file sharing software, such as BitTorrent, Napster, Kazaa, and Gnutella. DHT research was made popularby the CAN, Chord, Pastry, and Tapestry projects. BitTorrent is the most scalable and fastest network, but the issue with BitTorrent and the others is that there is no incentive for users to keep the files indefinitely. Users generally don't keep files permanently, and if nodes that have data still required by someone leave the network, there is no way to retrieve it except by having the required nodes rejoin the network so that the files once again become available.

Two primary requirements here are high availability and link stability, which means that data should be availablewhen required and network links also should always be accessible. **Inter-Planetary File System** (**IPFS**) by JuanBenet possesses both of these properties, and its vision is to provide a decentralized World Wide Web by replacing the HTTP protocol. IPFS uses Kademlia DHT and Merkle **Directed Acyclic Graphs** (**DAGs**) to provide storage and searching functionality, respectively.

The incentive mechanism for storing data is based on a protocol known as Filecoin, which pays incentives to nodes that store data using the Bitswap mechanism. The Bitswap mechanism lets nodes keep a simple ledger of bytes sent or bytes received in a one-to-one relationship. Also, a Git-based version control mechanism is used inIPFS to provide structure and control over the versioning of data.

There are other alternatives for data storage, such as Ethereum Swarm, Storj, and MaidSafe. Ethereum has its own decentralized and distributed ecosystem that uses Swarm for storage and the Whisper protocol for communication. MaidSafe aims to provide a decentralized World Wide Web.

BigChainDB is another storage layer decentralization project aimed at providing a scalable, fast, and linearly scalable decentralized database as opposed to a traditional filesystem. BigChainDB complements decentralizedprocessing platforms and filesystems such as Ethereum and IPFS.

Communication

The Internet (the communication layer in blockchain) is considered to be decentralized. This belief is correct to some extent, as the original vision of the Internet was to develop a decentralized communications system. Services such as email and online storage are now all based on a paradigm where the service provider is in control, and users trust such providers to grant them access to the service as requested. This model is based on the unconditional trust of a central authority (the service provider) where users are not in control of their data. Even user passwords are stored on trusted third-party systems. Thus, there is a need to provide control to individual users in such a way that access to their data is guaranteedand is not dependent on a single third party.

Access to the Internet (the communication layer) is based on **Internet Service Providers** (**ISPs**) who act as a central hub for Internet users. If the ISP is shut down for any reason, then no communication is possible with thismodel.

An alternative is to use **mesh networks**. Even though they are limited in functionality when compared to the Internet, they still provide a decentralized alternative where nodes can talk directly to each other without a central hub such as an ISP.

Now imagine a network that allows users to be in control of their communication; no one can shut it down for any reason. This could be the next step toward decentralizing communication networks in the blockchain ecosystem. It must be noted that this model may only be vital in a jurisdiction where the Internet is censored andcontrolled by the government.

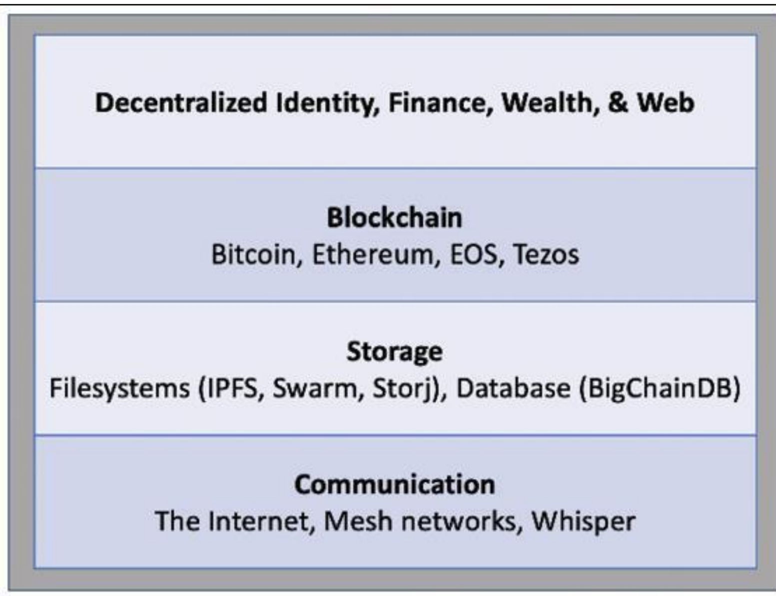
As mentioned earlier, the original vision of the Internet was to build a decentralized network; however, over the years, with the advent of large-scale service providers such as Google, Amazon, and eBay, control is shifting toward these big players. For example, email is a decentralized system at its core; that is, anyone can run an email server with minimal effort and can start sending and receiving emails. There are better alternatives available. For example, Gmail and Outlook already provide managed services for end users, so there is a natural inclination toward selecting one of these large centralized services as they are more convenient and free to use. This is one example that shows how the Internet has moved toward centralization.

Free services, however, are offered at the cost of exposing valuable personal data, and many users are unaware of this fact. Blockchain has revived the vision of decentralization across the world, and now concerted efforts are being made to harness this technology and take advantage of the benefits that it can provide.

Computing power and decentralization:

Decentralization of computing or processing power is achieved by a blockchain technology such as Ethereum, where smart contracts with embedded business logic can run on the blockchain network. Other blockchain technologies also provide similar processing-layer platforms, where business logic can run over the network in a decentralized manner.

The following diagram shows an overview of a decentralized ecosystem. In the bottom layer, the Internet or mesh networks provide a decentralized communication layer. In the next layer up, a storage layer uses technologies such as IPFS and BigChainDB to enable decentralization. Finally, in the next level up, you can see that the blockchain serves as a decentralized processing (computation) layer. Blockchain can, in a limited way, provide a storage layer too, but that severely hampers the speed and capacity of the system. Therefore, other solutions such as IPFS and BigChainDB are more suitable for storing large amounts of data in a decentralized way. The Identity and Wealth layers are shown at the top level. Identity on the Internet is a vast topic, and systems such as bitAuth and OpenID provide authentication and identification services with varying degrees of decentralization and security assumptions:



Decentralized ecosystem

The blockchain is capable of providing solutions to various issues relating to decentralization. A concept relevantto identity known as **Zooko's Triangle** requires that the naming system in a network protocol is secure, decentralized, and able to provide human-meaningful and memorable names to the users. Conjecture has it that a system can have only two of these properties simultaneously.

Nevertheless, with the advent of blockchain in the form of **Namecoin**, this problem was resolved. It is now possible to achieve security, decentralization, and human-meaningful names with the Namecoin blockchain. However, this is not a panacea, and it comes with many challenges, such as reliance on users to store and maintain private keys securely. This opens up other general questions about the suitability of decentralization toa particular problem.

Decentralization may not be appropriate for every scenario. Centralized systems with well-established reputations tend to work better in many cases. For example, email platforms from reputable companies such as Google or Microsoft would provide a better service than a scenario where individual email serversare hosted by users on the Internet.

There are many projects underway that are developing solutions for a more comprehensive distributed blockchain system. For example, Swarm and Whisper are developed to provide decentralized storage and communication for Ethereum.